Accessible Legal Assistance

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Milestone 3 - Wrap Up

1. Methodology

This project develops a retrieval-augmented, verifiable, and explainable legal question-answering system designed to assist non-experts in understanding complex contractual language. The system uses a modular multi-agent architecture implemented through LangGraph, consisting of three specialized components: a Retriever Agent, a Reasoner Agent, and an Explainer Agent. Each agent plays a distinct role in the legal reasoning pipeline, enabling structured, citation-supported, and comprehensible responses.

A multi-agent retrieval-augmented generation pipeline was constructed on the CUAD (Hendrycks et al., 2021) corpus to provide verifiable, lay-accessible explanations of contract language. Contracts were loaded from CUAD, split into approximately 600-token segments with a 100-token overlap via a recursive splitter, and embedded using `BAAI/bge-large-en-v1.5` with automatic device selection (CPU/MPS/CUDA). Embeddings and metadata were indexed in FAISS for semantic search, while the raw documents were cached for BM25 keyword retrieval, enabling hybrid search. Hybrid retrieval combined FAISS and BM25 results through Reciprocal Rank Fusion to balance semantic recall and lexical precision; an optional cross-encoder reranker (`ms-marco-MiniLM-L-12-v2`) re-ranked the top candidates when enabled. An LLM-driven query rewrite loop improved recall by expanding legal terminology and synonyms; up to two rewrites were allowed before proceeding.

The agent workflow (LangGraph) routed each query through retrieval and a relevance grader (structured yes/no) to filter weak matches; low-relevance paths triggered rewriting, while high-relevance paths advanced to reasoning. The Reasoner operated under a strict citation regime: every sentence begins with `[Document X]`, and missing evidence is labeled “INSUFFICIENT EVIDENCE.” A Verifier (structured output) scored citation quality from 0–10, flagged missing citations, and detected advisory tone, providing a gate before explanation. The Explainer converted the reasoning into 8th-grade, citation-preserving prose with explicit educational disclaimers. Prompting was kept deterministic (temperature 0) to ensure repeatability.

Evaluation scripts executed fixed legal queries through the full pipeline and computed grounding (citation rate, unsupported-claim rate, grounding score), readability (average sentence length, average word length, readability score), and retrieval metrics, such as precision, recall, F1, and MRR, when ground truth was supplied. Results were aggregated via `EvaluationRunner`, and weights in code assigned 0.4 to grounding, 0.3 to readability, and 0.3 to retrieval when available (normalized otherwise).

2. Results

Across eleven legal queries, overall scores clustered tightly in the mid-0.7 range (mean 0.714; range [0.661, 0.751]). Grounding averaged 0.530, reflecting citation rates between 0.500 and 0.667 and unsupported-claim rates between 0.083 and 0.300. Readability remained consistently high (mean 0.960), with average sentence lengths from 13.4 to 21.8 words and average word lengths around 5.0–5.5 characters; readability scores stayed at or above 0.912 in every case.

The strongest overall performance appeared on "What are my options if the other party breaches the contract?" (0.751, grounding 0.579, readability 0.982) and "The service agreement has a limitation of liability clause" (0.740, grounding 0.611, readability 0.912). Termination clauses and indemnification queries also performed well (both 0.737 overall; grounding 0.540; readability 1.000), indicating stable retrieval and fluent explanation for common employment and liability scenarios. Vendor payment terms (0.743 overall; grounding 0.579; readability 0.963) benefited from relatively strong citation coverage (0.636) and low unsupported claims (0.091).

Weaker grounding emerged on more nuanced clauses. The non-compete query had the lowest overall score (0.661) and grounding (0.446) due to a lower citation rate (0.545) and higher unsupported-claim rate (0.182). The force majeure query showed similar grounding pressure (0.521; citation 0.636; unsupported 0.182). The most pronounced deficit appeared in "Can I terminate immediately if payment is late?" where unsupported claims rose to 0.300 and grounding fell to 0.420 despite a solid citation rate of 0.600, suggesting that critical assertions were not fully supported by the retrieved evidence. The confidentiality duration query (overall 0.735; grounding 0.579) and the NDA scope query (overall 0.670; grounding 0.438) illustrate how ambiguity in clause duration and scope can reduce grounding even when readability remains high (0.944–0.980).

All runs retrieved five documents and produced 8–12 reasoning steps, indicating consistent pipeline depth. High readability across every query, including those with weaker grounding, suggests that explanation quality remains stable while evidence alignment varies. Patterns across the set show that citation coverage is the main driver of grounding variance: queries involving conditional rights (immediate termination for late payment) and restrictive covenants (non-compete, force majeure) exhibited higher unsupported-claim rates and lower grounding scores, whereas more standard clauses (payment terms, limitation of liability, general termination) maintained stronger evidence alignment. These observations point to clause-specific difficulty rather than systemic failures in retrieval or explanation.

3. Analysis

The evaluation indicates that overall performance is stable (mean 0.714) with consistently high readability (mean 0.960), but grounding limits remain the primary constraint (mean 0.530). Citation coverage varied from 0.500 to 0.667 and unsupported-claim rates spanned 0.083–0.300, showing that a third to half of reasoning steps often lacked citations or contained partially supported statements. This variability clustered on nuanced or conditional clauses—non-compete scope, force majeure coverage, and immediate termination for late payment—where retrieved evidence was either sparse or only partially aligned to the claims being made. In contrast, standard clauses (payment terms, limitation of liability, general termination, indemnification) maintained stronger grounding, suggesting that both retrieval and reasoning prompts are better calibrated for common patterns in CUAD.

High readability across all queries, including those with weaker grounding, shows that explanation fluency is decoupled from evidence alignment; the Explainer maintains clear, low-complexity prose even when citations are incomplete. Consistent retrieval depth (five documents; 8–12 reasoning steps) suggests that grounding gaps are driven more by evidence selection and citation discipline than by pipeline truncation. The strict citation prompt and Verifier reduced hallucinations compared to earlier baselines, yet citation rates below full coverage indicate remaining gaps: either retrieval did not surface narrowly relevant passages, or the Reasoner failed to cite available evidence on every step. Unsupported-claim spikes (up to 0.300) on conditional rights imply that the model inferred beyond the text when contractual contingencies were not explicit.

Overall, the system is strong on readability and maintains mid-0.7 overall scores, but grounding is the leverage point. Improving citation completeness (per-step enforcement), tightening retrieval-to-reasoning alignment (reranker on by default; more aggressive rewrites for ambiguous clauses), and adding clause-specific exemplars should reduce unsupported claims. Annotated ground truth for retrieval would further separate retrieval errors from reasoning/citation lapses and guide targeted improvements.

4. Conclusions

The system achieves stable mid-0.7 overall scores with consistently high readability, showing that the explanation layer remains clear and accessible across diverse contract queries. Hybrid retrieval, optional reranking, and structured multi-agent prompting (grader, rewriter, reasoner, verifier, explainer) collectively reduce hallucination relative to simpler RAG baselines and keep outputs deterministic. However, grounding remains the central limitation: citation coverage leaves many reasoning steps uncited in several queries, and unsupported-claim rates reach 0.300 on edge cases involving conditional or ambiguous clauses (non-compete scope, force majeure events, immediate termination for late payment). This indicates either retrieval misalignment or incomplete citation discipline by the Reasoner, even when relevant passages are retrieved.

Operational constraints persist: reliance on external APIs, additional latency and dependency when the reranker is enabled, a two-rewrite cap that may constrain recall on difficult queries, CUAD-centric data, synthetic evaluation queries without human adjudication, and unmodeled jurisdictional nuances. To address these gaps, the next phase should enforce per-step citations more strictly (potentially with rejection sampling or verifier gating), run reranking by default for harder clauses, and expand query rewriting for ambiguous terms. Clause-specific exemplars and annotated retrieval ground truth would help separate retrieval errors from reasoning lapses and guide targeted improvements. Longer-term steps include domain-adaptive fine-tuning (e.g., LoRA), reranker caching or quantization to reduce latency, PII and jurisdictional guardrails, and adaptive routing that tightens or relaxes rewrites based on grounding feedback.

5. Workflow

Contributions

Hyemin	Set up the retrieving layer
Hyunkyu	Implemented complete reasoning and explaining layer
	Orchestrated all layers
	Evaluated the model functionality and performance
	Developed the UI of the chatbot
Zalopher	Connected dataset to to retrieving layer
	Analyzed the results of the model

Weekly Plan

Date	Planned Tasks and Deliverables
Oct 12, 2025	~~Milestone 1 Submission~~
Oct 19, 2025	~~Revise proposal based on feedback~~
	~~Finalize dataset selection and preprocessing plan~~
Oct 26, 2025	~~Implement RAG retrieval module~~
Nov 02, 2025	~~Develop multi-agent reasoning chain using ReAct~~
Nov 09, 2025	~~Evaluate and revise: retrieval and reasoning layer~~
Nov 16, 2025	~~Milestone 2 Submission (Everyone)~~
Nov 23, 2025	~~Fine-tune explanation model with LoRA (Zalopher)~~
	~~Integrate explainability layer (Hyemin, Hyunkyu)~~
Nov 30, 2025	~~Evaluate and revise: reasoning verification and explainability (Everyone)~~
Dec 07, 2025	~~Prepare final documentation (Everyone)~~
Dec 10, 2025	~~Milestone 3 Submission (Everyone)~~
Dec 11, 2025	Final Presentation (Everyone)

References

Dan Hendrycks, Collin Burns, Anya Chen, and Spencer Ball. Cuad: An expert-annotated nlp dataset for legal contract review. *NeurIPS*, 2021.

Appendix A. GitHub Repository

https://github.com/hyunkyuu/accessible-legal-assistant